

IWAMIZAWA, Japan

WMO: 474130

Lat: 43.2117N

Lon: 141.7858E

Elev: 169

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.8	11.0	2.3	6.2	9.6	5.2	7.2	12.6	21.4	30.9	19.8	29.3	6.3	50	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.7	84.3	72.9	81.5	71.3	78.9	69.7	75.1	81.3	73.5	78.9	72.0	76.7	9.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	123.9	78.6	71.8	118.1	77.0	70.2	112.0	75.5	38.8	81.5	37.4	79.3	35.9	76.9	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.8	18.5	16.4	DB	1.1	89.2	4.1	2.4	-1.8	90.9	-4.2	92.3	-6.5	93.6	-9.5	95.4	(4)
			WB	0.5	77.1	4.0	2.0	-2.4	78.5	-4.8	79.6	-7.0	80.7	-10.0	82.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	21.8	23.3	31.0	43.0	54.3	62.0	69.4	71.4	63.9	51.6	38.9	27.0	(6)
(7)		DBStd	18.31	5.75	6.19	6.02	6.39	5.77	5.31	4.82	4.84	5.74	5.88	7.26	6.21	(7)
(8)		HDD50	3571	874	748	590	228	25	1	0	0	0	52	341	712	(8)
(9)		HDD65	7164	1339	1168	1055	660	335	118	16	7	89	416	784	1177	(9)
(10)		CDD50	2324	0	0	0	18	157	361	602	664	416	100	6	0	(10)
(11)		CDD65	442	0	0	0	0	2	28	152	205	55	0	0	0	(11)
(12)		CDH74	2135	0	0	0	0	35	155	678	1045	222	0	0	0	(12)
(13)		CDH80	405	0	0	0	0	5	19	122	233	26	0	0	0	(13)
(14)	Wind	WSAvg	7.5	7.1	7.5	7.9	8.4	8.4	7.3	7.3	7.0	6.8	7.3	7.8	7.3	(14)
(15)	Precipitation	PrecAvg	48.6	4.5	3.3	2.3	2.4	3.4	2.8	4.6	5.9	5.9	4.2	4.3	4.9	(15)
(16)		PrecMax	65.4	7.6	6.3	5.3	5.8	5.9	8.1	10.0	13.5	11.3	9.3	7.6	8.6	(16)
(17)		PrecMin	36.7	2.3	1.3	0.8	0.4	1.6	0.9	1.7	1.9	2.4	1.9	2.2	2.9	(17)
(18)		PrecStd	7.4	1.3	1.4	0.9	1.3	1.1	1.3	2.2	2.6	2.6	1.5	1.4	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	40.1	42.8	51.5	69.9	78.1	82.5	86.5	89.3	83.4	70.4	61.5	49.0	(19)	
(20)		MCWB	37.0	38.5	42.9	53.5	60.2	66.4	74.3	76.4	72.6	60.3	54.4	45.2	(20)	
(21)		DB	36.1	39.0	46.7	63.7	72.8	78.2	83.4	85.6	79.2	66.9	57.4	43.9	(21)	
(22)		MCWB	33.2	34.8	40.5	51.0	58.0	65.3	72.3	74.4	69.3	59.2	51.6	40.9	(22)	
(23)		DB	33.2	36.2	43.5	58.5	68.8	75.0	80.6	82.3	76.4	64.4	54.2	39.7	(23)	
(24)		MCWB	30.7	32.9	38.5	48.1	56.2	64.2	71.2	72.5	67.8	56.9	49.2	36.7	(24)	
(25)		DB	30.8	33.7	40.6	54.3	65.3	71.8	77.8	79.7	73.7	61.8	51.1	36.4	(25)	
(26)		MCWB	28.5	30.7	36.1	45.8	54.7	62.7	69.9	71.2	66.3	55.5	46.6	33.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	37.3	38.8	44.6	55.9	62.7	69.8	76.4	77.9	74.6	63.8	55.6	45.9	(27)	
(28)		MCDWB	40.0	41.8	48.5	65.2	73.0	77.9	83.1	86.5	80.7	67.7	60.1	48.6	(28)	
(29)		WB	33.7	35.6	41.7	52.5	60.2	67.2	74.5	75.8	72.1	60.8	52.4	41.2	(29)	
(30)		MCDWB	35.6	37.9	46.0	61.1	68.6	74.7	80.3	82.7	76.9	65.3	56.6	43.8	(30)	
(31)		WB	31.2	33.3	38.7	49.5	58.2	65.5	72.8	74.4	70.1	58.5	49.6	36.8	(31)	
(32)		MCDWB	33.0	36.1	42.6	57.2	65.8	72.6	78.1	80.1	73.9	62.9	53.5	39.3	(32)	
(33)		WB	28.6	30.7	36.6	46.8	56.3	63.7	71.3	72.9	67.6	56.3	46.8	33.6	(33)	
(34)		MCDWB	30.6	33.5	40.3	53.2	63.5	70.2	76.2	77.8	72.3	60.9	50.7	35.9	(34)	
(35)	Mean Daily Temperature Range	MDBR	11.6	12.9	13.0	16.3	17.7	16.1	13.9	13.7	15.8	15.7	12.2	10.4	(35)	
(36)		5% DB	13.4	15.2	16.2	23.6	24.3	21.2	17.8	17.0	17.3	18.1	17.7	15.0	(36)	
(37)		MCWBR	11.9	12.8	11.9	13.2	12.4	10.5	8.3	7.8	8.8	11.4	13.6	13.1	(37)	
(38)		5% WB	13.5	15.0	15.3	21.3	20.4	17.7	14.4	14.7	13.9	15.8	17.0	14.7	(38)	
(39)		MCWBR	12.5	13.3	12.0	13.3	11.5	9.4	7.3	7.3	8.1	11.4	14.2	13.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.342	0.394	0.463	0.523	0.538	0.523	0.507	0.473	0.416	0.408	0.390	0.357	(40)	
(41)		taud	1.975	1.828	1.737	1.775	1.843	1.959	2.071	2.182	2.294	2.172	2.111	1.994	(41)	
(42)		Ebn at Noon	234	240	239	237	239	243	246	250	256	238	218	214	(42)	
(43)		Edh at Noon	37	51	64	67	64	57	51	44	36	36	33	33	(43)	
(44)	All-Sky Solar Radiation	RadAvg	418	663	1045	1427	1628	1613	1486	1389	1160	814	460	338	(44)	
(45)		RadStd	35	59	93	146	132	148	166	100	99	49	47	31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page